# Project Title: Machine Learning Based Housing Price Prediction System

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# ABSTRACT

This project presents an end-to-end machine learning application for predicting housing prices using the California Housing dataset obtained from `sklearn.datasets`. The dataset was systematically preprocessed and split into training (70%), validation (15%), and testing (15%) sets to ensure reproducible and unbiased evaluation.

Multiple learning tasks were formulated from the same dataset, including regression for predicting median house value and classification by categorizing house values into low, medium, and high classes. Various machine learning models were implemented and evaluated, including Simple and Multiple Linear Regression, Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and Neural Networks. Model performance was assessed using appropriate evaluation metrics such as Mean Squared Error (MSE), $R^2$ score, accuracy, precision, recall, and F1-score.

Additionally, unsupervised learning techniques such as K-Means clustering were applied to identify housing patterns, and Principal Component Analysis (PCA) was used for dimensionality reduction and visualization. The best-performing regression model was deployed using a Flask-based web application, enabling users to input housing features and obtain real-time house price predictions.

This project demonstrates the complete machine learning pipeline, from data preprocessing and model development to evaluation and real-world deployment.

# INTRODUCTION

Housing price prediction is a significant real-world application of data science and machine learning. Accurate estimation of house prices plays a crucial role in assisting buyers, sellers, investors, and real estate agencies in making informed financial decisions. Housing prices are influenced by multiple socio-economic and geographical factors such as median income, population density, housing age, number of rooms, and location coordinates.

With the rapid growth of data availability, traditional statistical methods are often insufficient to capture complex relationships among these variables. Machine learning techniques provide powerful tools to analyze large-scale datasets, uncover hidden patterns, and model non-linear relationships effectively. By training predictive models on historical housing data, it becomes possible to estimate property values with improved accuracy and reliability.

In this project, the California Housing dataset from `sklearn.datasets` is used to develop an end-to-end machine learning application. The study includes three learning tasks derived from the same dataset: regression for predicting median house values, classification by categorizing houses into low, medium, and high value groups, and clustering to identify natural groupings of housing regions. Furthermore, dimensionality reduction using Principal Component Analysis (PCA) is performed for visualization purposes. Finally, the best-performing regression model is deployed through a Flask-based web application that enables real-time prediction of housing prices.

# DATASET DESCRIPTION

The dataset utilized in this project is the **California Housing Prices dataset**, obtained from the `sklearn.datasets` library. The data originates from the 1990 California census and contains detailed information about various housing block groups across the state of California.

Each record in the dataset represents a housing block group, while each attribute corresponds to a specific factor that influences housing prices. The dataset comprises eight input features and one continuous target variable.

The input features include: **Median Income (MedInc)**, **Median House Age (HouseAge)**, **Average Number of Rooms per Household (AveRooms)**, **Average Number of Bedrooms per Household (AveBedrms)**, **Population of the Block Group (Population)**, **Average Household Occupancy (AveOccup)**, **Latitude**, and **Longitude**. These features capture both socio-economic conditions and geographical characteristics of the housing regions.

The target variable, **Median House Value (MedHouseVal)**, represents the median price of houses within each block group and is used for regression analysis. For the classification task, this continuous variable is discretized into three categories — **low-value**, **medium-value**, and **high-value housing regions** — based on percentile thresholds, enabling multi-class classification.

# DATA PREPROCESSING

Data preprocessing is a crucial step in machine learning, as it ensures that the dataset is clean, consistent, and suitable for effective model training. In this project, the California Housing dataset was loaded using the `sklearn.datasets` library and converted into a pandas DataFrame to facilitate analysis and manipulation.

The dataset was thoroughly examined for missing or null values across all features. It was observed that the dataset contained no missing entries; therefore, no imputation or data cleaning was required.

To maintain fairness and avoid biased model evaluation, the dataset was split into three distinct subsets: a **training set (70%)**, a **validation set (15%)**, and a **testing set (15%)**. A fixed `random_state` value was used during splitting to ensure reproducibility of results across different runs.

Feature scaling was performed using the **StandardScaler** technique to normalize the input features and bring them to a common scale. The scaler was fitted exclusively on the training data and subsequently applied to the validation and test sets. This approach prevents data leakage and ensures that the model does not gain information from unseen data during preprocessing.

# EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was conducted to gain insights into the distribution of features and to understand the relationships between different variables in the California Housing dataset. All visualizations were generated using only the training dataset to avoid information leakage.

A histogram of the **Median Income (MedInc)** feature was plotted to examine its distribution. The visualization showed that the majority of housing regions fall within moderate income levels, while regions with very high median income are comparatively less frequent. This indicates a right-skewed distribution of income across housing blocks.

A scatter plot between **Median Income** and **Median House Value** was created to observe their relationship. The plot revealed a clear positive correlation, suggesting that higher income levels are generally associated with higher house prices. This trend highlights the strong influence of income on housing values.

Additionally, a correlation heatmap was generated to analyze the relationships among all input features and the target variable. The heatmap demonstrated that **Median Income** has the strongest positive correlation with median house value. Other features such as average number of rooms also showed moderate correlation, while geographical features like latitude and longitude exhibited comparatively weaker relationships.

Overall, EDA provided valuable insights into the key factors influencing housing prices and helped guide model selection and interpretation.

# REGRESSION MODELS

## 5.1 Simple Linear Regression

Simple Linear Regression was used to predict the median house value using a single feature, which was Median Income. This model helps in understanding the direct relationship between income levels and housing prices.

The model was trained using the training dataset and evaluated on the testing dataset. The results showed that Median Income has a strong influence on house prices, as higher income regions generally have higher housing values. The model provided a reasonable prediction accuracy but was limited because it considered only one feature.

## 5.2 Multiple Linear Regression

Multiple Linear Regression was implemented using all available features in the dataset. This model considers the combined effect of socio-economic and geographical factors on housing prices.

The model was trained on the scaled training data and evaluated on the testing data using Mean Squared Error and $R^2$ score. The results showed improved prediction performance compared to Simple Linear Regression, as it utilized multiple features to capture complex relationships within the dataset.

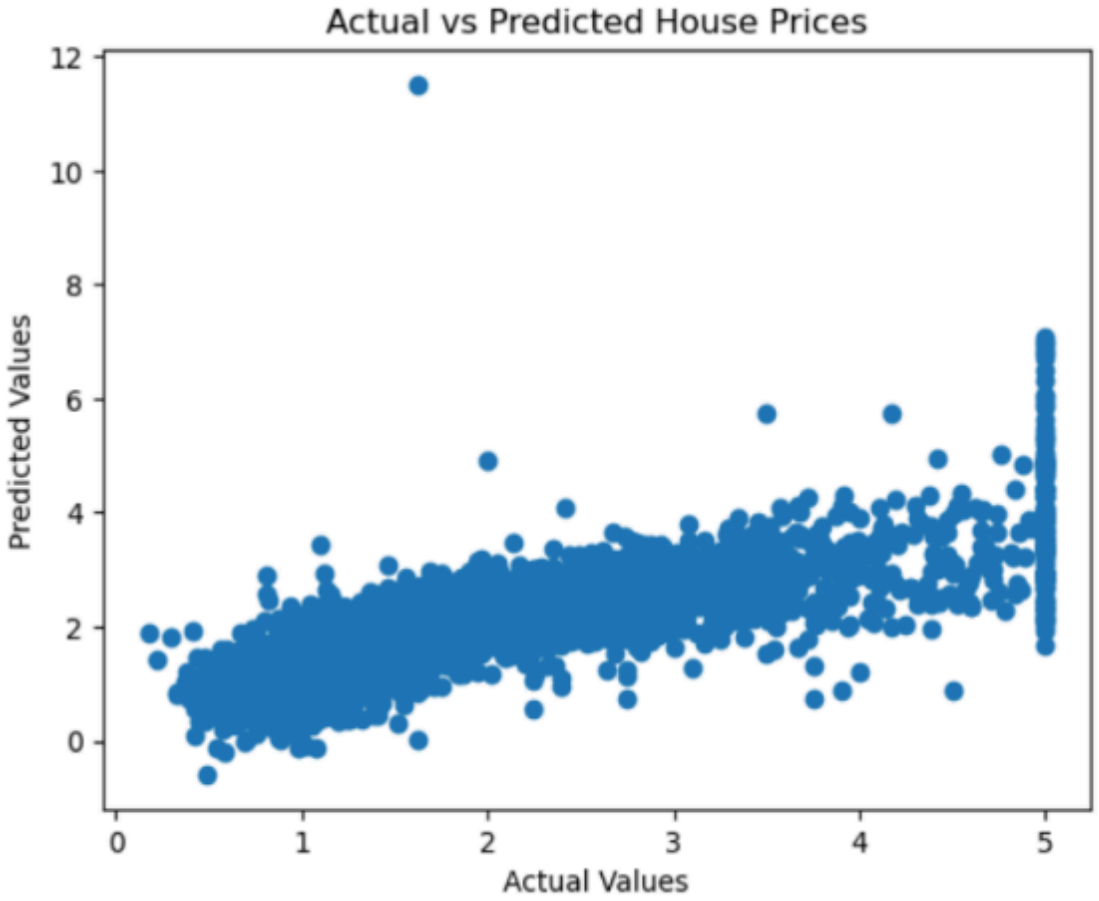


Figure 1: Actual vs Predicted values showing regression model performance.

# CLASSIFICATION MODELS

## 6.1 Logistic Regression

Logistic Regression was used as a classification model to categorize housing prices into three classes: low, medium, and high value houses. The target variable was converted into three categories based on percentile ranges.

The model was trained using the training dataset and evaluated on the test dataset. Logistic Regression provided a basic classification approach and demonstrated moderate accuracy in predicting housing value categories.

## 6.2 Decision Tree

Decision Tree classification was implemented to model complex relationships between housing features and price categories. The model works by splitting the dataset based on feature conditions to classify housing values into different categories.

The Decision Tree showed better performance compared to Logistic Regression because it can capture non-linear relationships between features and provides more accurate classification results.

## 6.3 Random Forest

Random Forest classification was applied as an ensemble learning method that combines multiple decision trees to improve prediction accuracy. This model reduces overfitting and provides more stable and reliable results.

Among all classification models used in this project, Random Forest achieved the highest accuracy, making it the best performing classifier for predicting housing value categories.

## Neural Network (MLP Classifier)

A Multi-Layer Perceptron (MLP) Neural Network was implemented to classify housing values into three categories. The neural network uses multiple hidden layers and activation functions to learn complex patterns from the dataset.

The model was trained using the scaled training dataset and tested on unseen data. It showed good classification performance and was able to capture nonlinear relationships between features more effectively than simpler models.

```
Neural Network Accuracy: 0.773578811369509
              precision    recall  f1-score   support

           0       0.84      0.81      0.83      1035
           1       0.65      0.73      0.69      1020
           2       0.85      0.78      0.81      1041

    accuracy                           0.77      3096
   macro avg       0.78      0.77      0.78      3096
weighted avg       0.78      0.77      0.78      3096
```

Figure 2: Neural Network classification performance results.

## Confusion Matrix and Classification Report:

```
Random Forest Accuracy: 0.8152454780361758
[[892 130  12]
 [142 747 131]
 [ 19 138 885]]
              precision    recall  f1-score   support

           0       0.85      0.86      0.85      1034
           1       0.74      0.73      0.73      1020
           2       0.86      0.85      0.86      1042

    accuracy                           0.82      3096
   macro avg       0.81      0.81      0.81      3096
weighted avg       0.82      0.82      0.82      3096
```

Figure 3: Performance of classification models showing confusion matrix and accuracy metrics.

## Comparative Discussion:

A comparative analysis of the classification models shows that Logistic Regression provided a basic baseline performance with moderate accuracy. The Decision Tree model improved classification results by capturing non-linear relationships in the dataset. However, the Random Forest classifier achieved the highest accuracy among all models due to its ensemble learning approach, which combines multiple decision trees to produce more stable and reliable predictions.

## 7. SUPPORT VECTOR MACHINE

Kernel Used: In this project, the Support Vector Machine model was implemented using the Radial Basis Function (RBF) kernel. The RBF kernel helps in handling non-linear relationships between features by mapping the data into a higher-dimensional space.

Results: The SVM model was trained on the training dataset and evaluated on the testing dataset. It achieved good classification accuracy and was able to effectively separate housing value categories based on feature patterns.

Comparison with Other Classifiers: When compared with other classification models, SVM performed better than Logistic Regression but slightly lower than the Random Forest classifier. Random Forest achieved higher accuracy due to its ensemble learning approach, which combines multiple decision trees to produce more stable predictions.

# 8. WEB DEPLOYMENT

## System Architecture:

The housing price prediction system was deployed as a web application using the Flask framework. The architecture consists of a trained machine learning model, a backend server implemented in Python using Flask, and a frontend interface built with HTML.

The trained model and scaler were saved using pickle files and loaded into the Flask application. The user inputs housing features through a web form, and the backend processes these inputs, scales them, and passes them to the trained model to generate predictions.

## Backend Implementation:

The backend of the system was implemented using Flask. The Flask server handles user requests, loads the trained model, processes input data, performs predictions, and sends results back to the frontend interface.

## Frontend Interface:

The frontend interface was created using HTML to provide a simple and user-friendly form where users can enter housing attributes such as income, house age, number of rooms, population, and location coordinates.

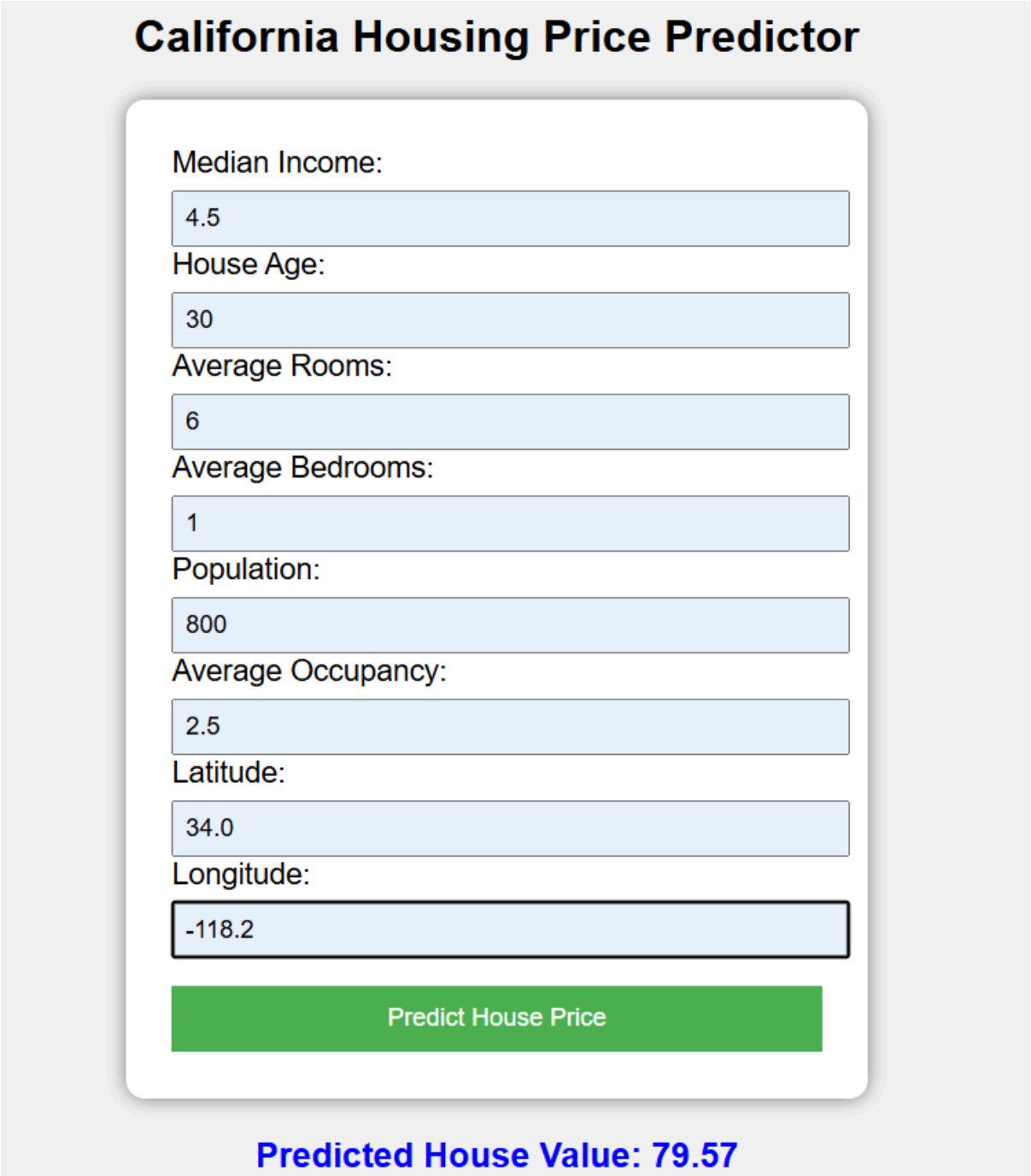


Figure 4: Web interface showing real-time housing price prediction using Flask application.

## 9. RESULTS & DISCUSSION:

The performance of different machine learning models was evaluated for regression, classification, and clustering tasks. In the regression analysis, **Multiple Linear Regression** performed better than **Simple Linear Regression**, as it utilized all input features and achieved higher prediction accuracy.

For classification, the **Random Forest** model achieved the highest accuracy, followed by **Support Vector Machine (SVM)** and **Neural Network**. **Logistic Regression** and **Decision Tree** showed moderate performance. The strong results of Random Forest highlight the effectiveness of ensemble learning in handling complex and non-linear relationships within the dataset.

In unsupervised learning, **K-Means clustering** successfully grouped housing regions based on socio-economic characteristics, while **Principal Component Analysis (PCA)** reduced dimensionality and improved data visualization. Overall, advanced machine learning models demonstrated better performance compared to simpler linear approaches.

## 10. CONCLUSION & FUTURE WORK:

This project successfully implemented an end-to-end machine learning system for predicting housing prices using the California Housing dataset. Regression, classification, clustering, and neural network models were developed and evaluated.

Multiple Linear Regression showed strong performance for price prediction, while Random Forest achieved the highest classification accuracy.

Future improvements may include using advanced deep learning models, integrating additional datasets, and deploying the system on cloud platforms for greater scalability.

## REFERENCES:


1. California Housing Dataset – sklearn.datasets

2. Scikit-learn Machine Learning Library Documentation

3. Flask Web Framework Documentation

4. Python Programming Language Documentation